

Application of Epimedium in Treatment of Erectile Dysfunction

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Abstract. Erectile Dysfunction (ED), means that Penis cannot reach or keep Erecting enough in sex to finish satisfactory sexual intercourse, is a common health issue of males. The prevalence of ED increase with the increasing of age of males. This review article first presents prevalence data for ED among Chinese males, noting that modern medical treatments (PDE5 inhibitors, vacuum erection devices) carry side effects or discomfort. It then analyses ED mechanisms from both Chinese and Western medical perspectives: Traditional Chinese Medicine (TCM) centers on "malnourished/lax tendons", while modern medicine categorizes it as organic ED, psychogenic ED, or a combination of both. Subsequently, it details the traditional application of Epimedium, documented in the Divine Farmer's Classic of Materia Medica for treating "impotence," its key active constituents (quercetin, kaempferol, icariin), and the pathways through which it improves ED. It also presents evidence-based clinical evidence of its use in formulations or proprietary Chinese medicines for ED intervention. Finally, it analyses current research limitations (lack of high-quality clinical evidence, risks of hepatic injury and endocrine effects, poor water solubility of icariin) and proposes future directions: deepening mechanism research, strictly controlling dosage and standardizing processing techniques, and leveraging nanotechnology to enhance quality, thereby providing reference for its clinical translation.

Keywords: Erectile Dysfunction, Epimedium, Traditional Chinese Medicine.

1. Introduction

Erectile Dysfunction(ED), a type of sexual dysfunction of males. The characteristic of ED is that the penis fails to achieve sufficient rigidity for vaginal penetration during sexual intercourse, or exhibits a marked reduction in firmness, such as becoming flaccid after erection and unable to sustain sexual activity. Even when an erection is achieved, it is difficult to maintain it until the conclusion of sexual intercourse, with frequent instances of mid-act flaccidity leading to interruption of the act. Or under the premise of sexual stimulation (Visual and Tactile stimuli), the penis unable to erect successfully, the time required to achieve an erection has significantly increased resulting in the inability to commence sexual intercourse normally. Apart from that, some patients always get psychological issues, for example, the feelings of inferiority, anxiety, loss of confidence in sexual relations, and even avoidance of intimate contact leaded by ED. If the ED is caused by basic disease like Diabetes or Hypertension, some symptoms associated with that condition may also be present. With the aging of population, ED is becoming a major health issue of the society gradually. In China, the prevalence of ED in males is about 43.4%, 20.6% for under 40, 40.6% for aged 40 to 59, and 77.7% for 60 years old and above [1].

In modern medicine, there are many major treatments of ED, such as Pharmacotherapy, by improving blood supply to the corpus cavernosum of the penis through medication, thereby promoting erection, PDE5 inhibitors like Sildenafil, Tadalafil and Vardenafil are used for treating commonly. But there are also some limitations for Pharamacotherapy, for example, some adverse reactions happened, part of patients said that they got headaches, facial flushing, visual disturbances and indigestions. This may affect patient's compliance of medications. Another method is using Vacuum Erection Device (VED) to help patients to induce penile erection through physical means, but the feedback of patients told us that the treatment of setting VED in penis may lead to additional discomforts like penisache and penis congestion etc.

The history of TCM in treating ED stretches over two millennia, the Huangdi Neijing record similar conditions such as "inability of male organ" and "laxity of Zongjin (the muscles in the lower

parts of male body)”. Shen Nong’s herbal classic lists dozens of medicinal substances used for the treatment of ED, among Epimedium is one. However, there remains no systematic or detailed review and analysis of the correlation between Epimedium and ED, nor of its specific applications.

Therefore, this paper aims to examine the core components and mechanisms of action of Epimedium in treating ED, focusing on pathogenesis of ED and integrating theories from both TCM and Modern Medicine. It will dissect the limitations of current research, and outline future directions for optimisation, thereby providing reference for its clinical translation and application.

2. Etiology and Mechanisms of Erectile Dysfunction

2.1. Pathogenesis and Pathomechanism in Traditional Chinese Medicine(TCM)

TCM posits that the aetiology of Erectile Dysfunction involves multiple organs and multifactorial causes. The penis is termed the “origene of the Zongjin” in TCM, being pivotal to Erectile Function. So the underlying pathogenesis of Erectile Dysfunction may be summarized as “Zongjin loss nutrition” and “flaccidity of Zongjin”.

Common aetiologies and pathogenesis of ED include deficiency of Kidney Essence, insufficiency of Kidney Yang, stagnation of Qi of liver, deficiency of both heart and spleen, descent of damp heat.

Deficiency of Kidney Essence arising from congenital deficiency or acquired depletion (such as excessive indulgence in sensual pleasures or age-related physical decline) fails to nourish the vital tendons, resulting in ED, night sweats (perspiration occurring during sleep that ceases upon waking), heat in the palms and soles, and dryness of the mouth and throat.

Insufficiency of Kidney Yang: Kidney Yang constitutes the foundation of the body's yang energy. When Kidney Yang is deficient, its warming and nourishing function becomes inadequate, failing to adequately sustain the genital tendons. This manifests as weak erections, aversion to cold, cold pain in the lower back and knees, frequent urination at night, and lethargy.

Liver Qi Stagnation: Emotional distress (anxiety, depression, etc.) impedes the smooth flow of liver Qi, obstructing the circulation of Qi and blood. Consequently, the genital tendons fail to receive adequate nourishment from qi and blood, manifesting as erectile dysfunction that fluctuates with emotional states (worsening after anger, improving when relaxed). Symptoms include chest and flank distension and pain, irritability, and quick temper.

Deficiency of heart and spleen: Excessive mental preoccupation and physical exertion deplete heart blood and impair spleen Qi, leading to insufficient Qi and blood production. The tendons lack Qi and blood support, often accompanied by poor appetite, palpitations, insomnia, pallor, fatigue, and loose stools.

Damp-heat descending: Consumption of pungent and greasy foods, prolonged sitting leading to internal generation of damp-heat, or external invasion by damp-heat pathogens descending into the liver meridian (which traverses the penis), obstructing the tendons of the genitalia, resulting in erectile dysfunction. This is often accompanied by scrotal dampness, yellowish-red urine, and a bitter, sticky sensation in the mouth.

2.2. Etiology and Pathogenesis in Modern Medicine

Modern medicine classifies Erectile Dysfunction into organic ED and psychogenic ED. The two types exhibit distinct characteristics in onset. Organic ED typically develops gradually in daily life, manifesting erectile difficulties across all scenarios—including masturbation and morning erections—and is often accompanied by diminished libido and abnormal ejaculation. Psychogenic ED frequently manifests abruptly, with no underlying physiological damage to the penis. Erection status is often contingent upon fluctuating psychological states (e.g., achieving erection when alone but failing during sexual intercourse), and is frequently associated with specific scenarios or emotional states such as anxiety or tension.

The aetiology of ED encompasses multiple factors. Organic ED arises from abnormalities in bodily organs or physiological functions, including vascular, neurological, endocrine, medication-related,

and penile structural factors [2]. Regarding vascular factors, the incidence of ED increases with the number of vascular comorbidities. Cardiovascular diseases such as hypertension, diabetes, hyperlipidaemia, and atherosclerosis can cause arterial ED by impairing arterial blood supply to the corpus cavernosum. Penile venous leakage (where blood cannot be effectively retained and rapidly refluxes) leads to insufficient rigidity or mid-erection softening. Neurologically, events damaging the central nervous network or peripheral nerves associated with sexual function can cause ED. Nerve injury disrupts the transmission of erection responses triggered by stimulating signals. Examples include diabetic neuropathy, lumbar disc herniation compressing nerves, damage to pelvic nerves during prostate surgery, and spinal cord injury. Endocrine (hormonal) factors: Most patients experience reduced libido and diminished vascular responsiveness to sexual stimuli due to low androgen (particularly testosterone) secretion levels, leading to ED. Other conditions such as hyperthyroidism, hypothyroidism, or hyperprolactinemia may also cause erectile dysfunction. Regarding medications, certain antihypertensive drugs, hormonal agents (anti-androgens), and psychotropic substances may reduce secretion of testosterone and other androgens. This can cause issues analogous to those arising from endocrine system-related ED, directly or indirectly affecting male erectile function [3]. At the anatomical level, structural abnormalities of the penis may also impede erection. For instance, conditions such as penile curvature deformity (Peyronie's disease), and congenital genital anomalies can create obstructions in the pathways through which blood is pumped into the corpus cavernosum via the arteries.

Psychological factors constitute the aetiology of psychogenic ED, with core triggers including prolonged anxiety and tension, or sudden significant psychological stress; Communication breakdowns with partners, emotional conflicts, and trust crises leading to psychological resistance or aversion towards sexual activity; Negative perceptions of sexual intercourse, such as fear of failure, excessive preoccupation with performance during sex, or unrealistic expectations regarding sexual encounters; Past unpleasant sexual experiences, lack of sexual knowledge, or cognitive biases resulting in psychological burdens.

Evidence-based data indicates that among the aetiological distribution of ED patients, 726 cases (62.6%) presented with purely psychogenic ED, 84 cases (7.2%) with purely organic ED, and 350 cases (30.2%) with coexisting psychogenic and organic factors. Among the latter two categories, vascular ED accounted for 297 cases (68.4%), comprising 175 venous and 122 arterial cases. Endocrine abnormalities caused 55 cases, drug-induced factors 16 cases, neurogenic factors 21 cases, and multiple confounding factors 45 cases [4].

3. The Mechanism of Action of Epimedium in Treating Erectile Dysfunction

The Shen Nong's herbal classic records: "Epimedium has a pungent flavour and cold nature. It treats impotence, treats injuries, promotes urination, enhances vigour, and strengthens the will." The Compendium of Materia Medica states: "Epimedium possesses a cold nature, inheriting the water energy of winter's celestial season, entering the Foot Shaoyin Kidney Meridian. Its pungent flavour is non-toxic, acquiring the moisturising metal quality of the earth, entering the Lung Meridian of the Hand Taiyin. Its nature descends more than it ascends, being yin. Yin governs the tendons; when water fails to restrain fire, excessive heat causes the tendons to lose their rigidity, leading to impotence. Epimedium enters the kidney with its cold nature; this cold sufficiently restrains fire, thereby curing impotence. In cases of complete injury, where yin is severed and essence is damaged, its cold nature nourishes water, while its pungent flavour provides moisture. This moisture replenishes yin essence." Modern medical research indicates that the primary active components in Epimedium responsible for treating ED include quercetin, kaempferol, and icariin glycosides.

Quercetin promotes the release of prostaglandin and nitric oxide synthase, reduces vascular endothelial apoptosis, enhances eNOS enzyme activity and penile cavernosal pressure, thereby improving penile cavernosal erectile function. This partially mirrors the mechanism of action of sildenafil [5].

Syneprhine activates thyroid hormones (TH), and studies indicate that thyroid hormones enhance the contractile activity of the corpus cavernosum of the penis. TH increases sympathetic nerve activity by influencing adrenergic receptors, with lumbar spinal cord-innervated postganglionic fibres of the lumbar sympathetic ganglion regulating penile erectile function [6]. The thyroid hormone signalling pathway enables TH to regulate the synthesis and secretion of steroid hormones via the endocrine axis, thereby influencing male sexual function [5].

Icariin may relax the smooth muscle of the penile corpus cavernosum to improve cavernosal blood flow and restore erectile function [5]. Icariin enhances nerve conduction and promotes nerve regeneration, exhibiting dual neuroprotective and neurotrophic effects while improving erectile function [7]. Research has also indicated that icariin may reduce oxidative stress or mitochondrial autophagy, thereby promoting erectile function [8]. Research through animal experiments has confirmed that icariin can dose-dependently reduce proteinuria, serum creatinine, and blood urea nitrogen levels in rats subjected to renal ischaemia-reperfusion injury. Concurrently, it enhances superoxide dismutase (SOD) and glutathione (GSH) activity while decreasing malondialdehyde (MDA) production. Mechanistically, icariin inhibits oxidative stress-induced apoptosis by upregulating the p-Nrf2/Nrf2 ratio and HO-1 protein expression (reducing Caspase-3 and Bax/Bcl-2 ratio), mitigating oxygen radical damage to vascular endothelium. Concurrently, its anti-inflammatory effects alleviate chronic inflammation in cavernous tissue, thereby preserving normal erectile physiology and reducing ED incidence [9]. Based on network pharmacology analysis, epimedium may exert its effects on ED by modulating 36 potential targets [10].

4. The Application of Epimedium in the Treatment of Erectile Dysfunction

TCM holds that Epimedium has a pungent flavour and cold nature, entering the Liver and Kidney meridians. Its core efficacy lies in supplementing Kidney yang, strengthening tendons and bones, and dispelling wind-dampness. For ED cases diagnosed under TCM pattern differentiation as "Kidney Yang Deficiency" (manifesting as weak erections, soreness in the lower back and knees, aversion to cold with cold limbs, and lethargy), Epimedium is frequently incorporated into compound herbal formulas. It works by "warming and tonifying Kidney Yang" to enhance overall bodily function, thereby indirectly supporting reproductive regulation rather than directly addressing ED through symptomatic treatment.

In traditional Chinese medicine, Epimedium is frequently combined with other herbal ingredients to form formulations for treating ED. For instance, the Compendium of Materia Medica records the "Xianlingpi Wine" (prepared by steeping Epimedium in white liquor, possessing kidney-tonifying and yang-strengthening effects, indicated for kidney yang deficiency-type ED); In contemporary clinical practice, it is combined with prepared rehmannia root, cornelian cherry, epimedium, cuscuta, astragalus, and codonopsis to address spleen-kidney deficiency-type ED. Over-the-counter preparations containing epimedium include Compound Xuanju Capsules and Sanzi Qiangshen Capsules. A meta-analysis encompassing 42 randomised controlled trials (n=3904) found that intervention groups uniformly received Compound Xuanju Capsules, while control groups received alternative treatments. Outcomes measured included the International Index of Erectile Function-5 (IIEF-5) or serum total testosterone (TT). Results demonstrated that Compound Xuanju Capsules significantly improved IIEF-5 scores (40 studies, MD = 3.68, 95% CI: 2.72 to 4.63, $P < 0.00001$); and markedly improved TT levels (8 studies, MD = 3.83, 95% CI: 1.84 to 5.83, $P = 0.0002$) [11]. Additionally, several studies have demonstrated that Shanhaidan Granules and Shugan Yiyang Capsules can significantly enhance erectile function and morning erections compared to placebo [12, 13].

5. Current Limitations of Treatment and Future Prospects

Among drugs currently developed according to modern theories, none contain Epimedium or its constituents such as icariin, indicating that research into ancient medicine remains insufficiently comprehensive and thorough. Epimedium is often combined with other Chinese herbal medicines for treating ED. Although it can be concluded that organic components within Epimedium, such as quercetin, kaempferol, Icariin indeed possess corresponding therapeutic targets for ED and demonstrate significant effects in promoting erection in rats. However, comprehensive clinical studies proving the significant efficacy of isolated Epimedium in ED treatment remain scarce. The therapeutic outcomes from the few existing clinical trials lack generalisability and are deficient in multicentre, double-blind, placebo-controlled design. Furthermore, practical application of epimedium in ED treatment carries potential side effects, such as liver damage arising from variations in processing techniques [14]. It may also exert certain effects on the endocrine system, such as promoting oestrogen secretion, leading to gynaecomastia and reduced sperm count [15].

Due to the poor water solubility of the active ingredient icariin, its therapeutic efficacy in practical treatment is negligible. Currently, the primary method of administering epimedium for treating erectile dysfunction is oral ingestion. The low oral bioavailability resulting from its poor water solubility is one of the limitations in its application [16].

In future treatment approaches, it is imperative to deepen research into molecular mechanisms and identify precise therapeutic targets. Further clinical trials are required to substantiate that Epimedium and its constituent compounds effectively ameliorate erectile dysfunction. To prevent hepatic impairment, the Chinese Clinical Application Guidelines for Traditional Chinese Medicines recommend that patients with kidney yang deficiency-type ED may use epimedium for short-term periods (≤ 1 month), with a daily total flavonoid dose not exceeding 100mg. In clinical ED treatment, epimedium requires in-depth mechanistic and clinical research, strict dosage control, and standardised processing techniques to mitigate its potential effects on liver function and endocrine systems. Regarding Epimedium's water solubility, nanomedicine technology may be employed to enhance its efficacy and safety profile [17].

6. Conclusion

ED is a prevalent condition significantly impacting men's quality of life, with a prevalence rate of 43.4% among Chinese males. This rate increases markedly with age, highlighting substantial clinical treatment demand. TCM and Western medicine offer complementary insights into the mechanisms of ED. TCM centres on concepts of "inadequate nourishment of the Zongjin" and "laxity of the Zongjin", analysing the condition through different syndromes such as deficiency of kidney essence and insufficiency of kidney yang. Modern medicine categorises ED into organic (involving vascular or neurological abnormalities), psychogenic (related to psychological stress or cognitive biases), and mixed types, providing a theoretical foundation for clinical intervention.

Epimedium is a classic medicinal herb traditionally used to treat erectile dysfunction, recognised by both traditional and modern medicine. The ancient Shen Nong's herbal classic of *Materia Medica* records its efficacy in treating impotence. Modern research has identified active components such as quercetin, kaempferol, and icariin, which improve erectile dysfunction through multiple pathways: regulating sympathetic nerve activity and hormone secretion; relaxing smooth muscle; and providing antioxidant effects. Clinically, it is commonly administered in formulations such as "Xianlingpi Wine" or proprietary Chinese medicines like Xuanju Capsules and Liver-Soothing Yang-Enhancing Capsules for ED treatment, all demonstrating significant improvements in erectile function.

However, the application of this herbal medicine remains limited due to factors such as insufficient clinical evidence and the lack of multicentre double-blind controlled studies; liver damage and endocrine effects resulting from improper processing; and the poor water solubility of its active component icariin, leading to low oral bioavailability. Future efforts should focus on deepening

mechanistic and clinical research, strictly adhering to treatment guidelines, standardising processing techniques, and enhancing component water solubility through nanotechnology.

In summary, Epimedium possesses unique value in ED treatment, yet further clinical research and technological innovation are required to overcome current limitations, thereby providing more reliable clinical options for managing ED.

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